How to indentify ↓ Dimensional space

x_1	x_2

① Standardize each of every feature $\left(z = \frac{x-\mu}{\sigma}\right)$

② find underline sub-space

$$P = \frac{uu^T}{u^Tu}$$

If we need to project $\bar{x}$ onto u

The location of x on $u = Px$

(subspace such is which gives magnitude of each point highest)

$$P_x = \frac{uu^T}{u^Tu}$$

(u is unit vector (dot $u^Tu = 1$) unit vector)

$$Px = uu^Tx$$

$$\boxed{Px = (x^Tu)u}$$

$$\underset{u}{\operatorname{argmax}} = \frac{1}{n}\sum_{i=1}^{n}\left\|(x^{(i)T}u)\,u\right\|_2^2$$

$$\frac{1}{n}\sum_{i=1}^{n}(x^{(i)T}u)^2$$

both are same calculated BY EF

$$= \frac{1}{n}\sum_{i=1}^{n}(x^{(i)T}u)(x^{(i)T}u)$$

L_2 Norm will give magnitude of vector from 0 how far it is

Distance of each of every point

$\frac{1}{n}$ is scaling factor

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Lecture 53: BDBH-204-ML

$v^T A u$

↳ covariance matrix

PCA

Numerical

1) Computing Correlation & Covariance

x: Hours studied

y: Exam Scores

X	Y
1	2
2	4
3	5
4	4
5	5

* Covariance & Correlation

$$\boxed{\text{Correlation } r = \frac{Cov(x,y)}{\sigma_x \sigma_y}}$$

Covariance

$$= \frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$$

ii) $\sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \bar{x})^2}$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x^{(i)}$$

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x	$x-\bar{x}$	$(x^{(i)}-\bar{x})^2$	y	$y-\bar{y}$	$(y-\bar{y})^2$
1	-2	4	2	-2	4
2	-1	1	4	0	0
3	0	0	5	1	1
4	1	1	4	0	
5	2	4	5	1	1
15	0	10	20	0	6

$\bar{x} = 3$ $\qquad$ $\bar{y} = 4$

$(x^{(i)}-\bar{x})(y-\bar{y})$

$-2 \times -2 = 4$

$-1 \times 0 = 0$

$0 \times 1 = 0$

$1 \times 0 = 0$

$2 \times 1 = 2$

$\frac{1}{5} \quad \frac{6}{1.2}$

$\sigma_x = 1.414$

$\sigma_y = 1.0954$

$\sqrt{\frac{1}{4}}$

$$\text{Correlation} = \frac{1.2}{1.41 \times 1.09} = 0.7807$$

* Eigenvalues & Eigenvector

$\det(A - \lambda I) = 0$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \qquad 4 - 1 = 3$$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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$$= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\begin{bmatrix} \lambda - 2 & 0 \\ 0 & \lambda - 2 \end{bmatrix} \quad (\lambda - 2)(\lambda - 2) - 0 = 0$$

$$\lambda^2 - \lambda - \lambda 2 - 4 = 0$$

$$\underline{\lambda^2 - \lambda 3 - 4} = 0$$

$$-\lambda 4 + \lambda \qquad -4$$

$$\cancel{\lambda 3 - 1}$$

$$\lambda_1 = 1 \qquad -\lambda = 4 \qquad \lambda_2 = 4$$

$$\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)(2-\lambda) - 1$$

$$\cancel{4 + \lambda^2 - 4\lambda}$$

$$4 - 2\lambda - 2\lambda + \lambda^2 - 1$$

$$\lambda^2 - 4\lambda + 3$$

$$\lambda^2 - 3\lambda - 1\lambda\, 3 =$$

$$\lambda_1 = 3 \qquad \lambda_3 = 1$$

$A - \lambda I = 0$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{x_1}{x_2} \quad \text{if} \Rightarrow \lambda = 3 \quad \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{x_1}{x_2}$$

PCA

$$X = \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix} \\ \begin{array}{ccc} 1.5 & 1.5 & 1.2 \end{array} \end{array} \qquad d = 3$$

Step: 1 Scale

$$\mu = [1.5, 1.5, 1.5]$$

$$X_c = \begin{bmatrix} 0.5 & -1.5 & -0.5 \\ -1.5 & 0.5 & -0.5 \\ 1.5 & -0.5 & 0.5 \\ -0.5 & 1.5 & 0.5 \end{bmatrix}$$

Step 2 Compute Covariance matrix

$$\begin{bmatrix} Cov(x_1 x_1) & Cov(x_1 x_2) & Cov(x_1 x_3) \\ Cov(x_2 x_1) & Cov(x_2 x_2) & Cov(x_2 x_3) \\ Cov(x_3 x_1) & Cov(x_3 x_2) & Cov(x_3 x_3) \end{bmatrix}$$

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$$C = \frac{1}{n} X_c^T X_c$$

$$= \begin{bmatrix} 1.25 & -0.75 & 0.5 \\ -0.75 & 1.25 & 0.5 \\ 0.5 & 0.5 & 0.25 \end{bmatrix}$$

Step 3 :- Eigen value & Eigenvectors

$\lambda_1 = 2.0$, $\lambda_2 \simeq 0.75$, $\lambda_3 \simeq 0.0$

$$\cancel{y_1 = \begin{bmatrix} 1 \\ \sqrt{2} \end{bmatrix}}$$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad u_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \quad u_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Dimensionality Reduction Take top 2 Vectors
3D → 2D

$$W_{2D} = [v_1 \ v_2]$$

Project the data Points

$$z = X_c W_{2D} = [\ ]$$

Each point will be of 2D

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3D → 2D

Taken top 1 eigen vector

$W_{1P} = [v1]$

Total Variance

$= 2 + 0.75 + 0$

$= 2.75$

$PC1 = \frac{2}{2.75} \simeq 73\%$

$PC2 = \frac{0.75}{2.75} \simeq 27\%$

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Lecture 54: BDBH-204-M

- clustering (K-Mean)

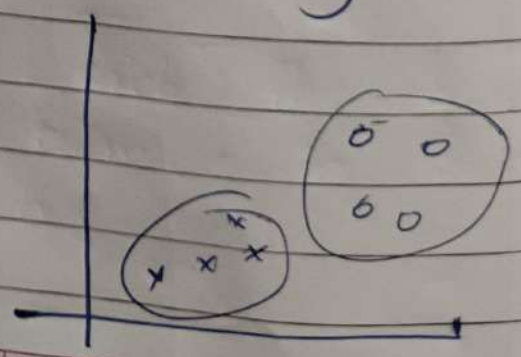
Total variance $\lambda_1 = 2.0$
$\lambda_2 = 0.75$
$\lambda_3 = 0$

$$u_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\\0\end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{6}}\begin{bmatrix}1\\1\\2\end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{3}}\begin{bmatrix}-1\\-1\\1\end{bmatrix}$$

$$PC1 = \frac{2}{2.75} \simeq 73\%$$

$$PC2 = \frac{0.75}{2.75} \simeq 27\%$$

$$PC3 = \frac{0}{2.75} \simeq 0$$

Unsupervised learning
1) Visualization → PCA, t-SNE
2) Clustering or Grouping
   a) K-means
   b) Hierchical clustring
      ↳ Agglomirative clustring

1) K-means Algorithm